

LiSA: Light-Space Atlases for Relightable Gaussian Splatting

Anonymous CVPR submission

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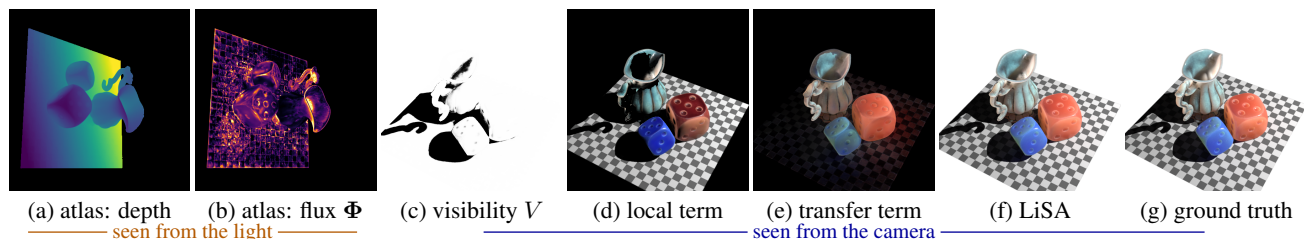


Figure 1. **Relighting from the light’s point of view.** One rasterization of the Gaussians from the point light records how deep the light is absorbed (a) and what it hits, through learned flux features (b, one of eight channels). A camera pixel reads this atlas to decide whether it is in shadow (c), how much light it reflects locally (d), and how much reaches it through the translucent vase and dice (e). The sum (f) matches the test view (g) at 34.4 dB, against 32.6 dB for the best of four recent baselines. Components (d, e) add in linear radiance and are shown on black.

Abstract

When a point light moves around an object, much of the change in its appearance comes from where the light lands: on the surfaces it reaches, away from those it misses, and through light that re-emerges elsewhere after entering translucent material. Relightable Gaussian splatting methods estimate these effects one primitive at a time. We argue that the light’s own image plane is a better place to reason about them. Because every pixel of a view from the light carries the same flux, the compositing weights obtained by rasterizing the Gaussians from the light are the fractions of light that each primitive intercepts. LiSA stores this deposit in a light-space atlas of depth moments and learned flux features, and every receiver reads a multi-scale pyramid of it to obtain a soft shadow test refined by a small network and a transfer term that is linear in the flux deposited around it. Making the model trainable also meant moving the loss from display to linear radiance while the geometry forms, which alone raises held-out PSNR by 6–7 dB on two scenes with thin, cluttered structures. On 18 scenes from three public benchmarks, LiSA reaches 33.47 dB average PSNR against 31.59 dB for the strongest of four recent baselines, with the best SSIM and LPIPS, and trains in about 20 minutes per scene.

1. Introduction

Photographing an object one light at a time (OLAT) and relighting it from new viewpoints is one of the oldest routes to photorealistic digital assets [10]. Its modern form reconstructs a relightable representation from hundreds to thousands of images, each lit by a single calibrated point light [2, 45], and Gaussian splatting [23] has made the result renderable in real time [4, 14, 25, 27, 49].

Under a point light, much of what makes an image look right is settled before any material is evaluated: which parts of the object the light reaches, which parts it misses, and how much of the light that enters the object leaves it somewhere else. Current Gaussian methods settle these questions one primitive at a time. GS³ [4] and SSD-GS [49] splat the scene toward the light to give every Gaussian one visibility value, RNG [14] reduces a depth shadow map to a scalar cue, and translucency is delegated to per-Gaussian networks conditioned on the local light direction [12, 49]. A large Gaussian on a floor then holds a single shadow value, so every moving shadow boundary has to be carved out by densification, and a network that sees only one primitive must memorize, light by light, how much light reaches it through the rest of the object.

We take a different vantage point, literally that of the light. Every pixel of a view rendered from a point light subtends

roughly the same solid angle and therefore carries the same flux. Rasterizing the Gaussians from the light with standard front-to-back alpha compositing divides that flux among the primitives along each ray, and within the splatting model the compositing weights are exactly the fractions of light that each Gaussian intercepts. A single extra rasterization thus records where light is absorbed, at what depth, and by what. Forward rendering has exploited the light's view for decades through shadow maps [13, 33, 42] and translucent [7] and reflective [8] shadow maps, with hand-designed quantities on known geometry. We bring it into inverse rendering, where the geometry and the stored quantity are both learned.

Our method, LiSA (Light-Space Atlas), splats for every light the first two depth moments of this deposit and a few learned flux features, and filters them into a multi-scale pyramid (Fig. 1). A receiver, either a Gaussian center or a pixel, projects into the light and reads the pyramid. The moments yield a soft, filtered shadow test in the spirit of variance shadow maps [13], which a residual network refines, and the flux features feed a transfer term that is linear in the light deposited around the receiver, a learned counterpart of translucent shadow maps. The atlas costs one 512^2 rasterization per image, its inputs are all relative to the light, and gradients flow from every camera pixel through it back into the Gaussians.

A good representation turned out not to be enough. Our first full-benchmark runs collapsed thin, cluttered structures such as the bucket of the Lego bulldozer into smooth blobs (Fig. 4), and variations of the initialization, densification, normal parameterization and material model all left Lego between 22 and 24 dB on held-out views. The culprit was the domain of the loss. Like many relighting pipelines, we compared images after the display transform, which weights every residual by the slope of the tone curve, and that slope grows without bound in the dark, shadowed regions that a light-space model has to explain. We therefore stage the optimization radiometrically: a network-free point-light model first fits the geometry while the target is annealed from display to linear radiance, as GS³ does for its HDR data [4]; the full model then trains in linear radiance on Gaussian receivers, so that densification sees ordinary rasterization gradients; and a final pass refines appearance per pixel on frozen geometry. With initialization, schedule and random state held fixed, changing only the loss domain raises held-out PSNR from 23.0 to 30.2 dB on Lego and from 24.1 to 30.1 dB on Drums.

On all 18 scenes of three public OLAT benchmarks [4, 12, 45], with GS³, SSD-GS, OLAT Gaussians [25] and RNG retrained under one protocol and scored against identical targets, LiSA attains the best average PSNR, SSIM and LPIPS (33.47 vs. 31.59 dB for the next best method), reproduces moving shadows that per-primitive methods fragment, and trains in about 20 minutes per scene. We contribute:

- a light-space formulation of point-light transport for Gaussian splats, in which one alpha-composited rasterization from the light records the light each primitive intercepts, yielding soft per-receiver visibility and a transfer term linear in the deposited flux;
- a radiometrically staged optimization that makes neural light-space shading trainable on difficult geometry, with a controlled study showing that the loss domain, not model capacity, caused the failures we observed;
- a unified evaluation of five methods on 18 scenes, multi-seed ablations, and an analysis of where the atlas transfer helps and where a local model suffices.

2. Related Work

Relighting from point-light captures. Neural reflectance fields recover relightable volumes from collocated or OLAT illumination [2, 3], and NRHints [45] conditions a radiance field on ray-traced shadow and highlight hints; related work factors scenes into geometry and BRDFs under environment light [21, 37, 48], including on Gaussians [5, 16, 17, 20, 26]. For point lights, GS³ [4] combines spatial and angular Gaussians with splatted shadows and a residual network, RNG [14] shades neural Gaussian features with a depth shadow-map cue, OLAT Gaussians [25] are guided by a proxy mesh from neural surface reconstruction [39], BiGS [27] uses bidirectional spherical harmonics, SSS-GS [12] adds a neural subsurface residual, and SSD-GS [49] separates scattering, shadows and reflection. RadiosityGS [19] and stochastic ray tracing [43] instead solve light transport explicitly on Gaussians, and relightable avatars learn transfer in a mesh or UV domain [35, 40] that general objects lack. All of these evaluate visibility or scattering per primitive or per ray; LiSA moves both into the light's image plane.

Visibility for Gaussians. Per-Gaussian visibility comes from splatting toward the light [4, 49], from tracing rays through Gaussians [16, 17, 31, 43], or from depth shadow maps [14]; deep Gaussian shadow maps [30] tabulate the transmittance of fixed Gaussians to let avatars and scenes shadow each other. Our atlas is instead filtered into a pyramid, evaluated at arbitrary receivers, and learned together with the geometry it describes.

Light-space rendering. Shadow maps [42] and their filtered variants [1, 13, 33, 34], deep and opacity shadow maps for semi-transparent media [24, 28], translucent shadow maps [7] built on diffusion models [11, 18], and reflective shadow maps [8] store per-light quantities in the light's view and gather them at shading time; precomputed radiance transfer makes outgoing light linear in the incident lighting [36], and multi-scale light representations gather indirect

light [6, 22]. Learned counterparts filter shadow-map buffers for known geometry [9] or decode rasterized buffers with networks [15, 32, 38, 47]. LiSA keeps the light-space gather and the linearity in incident flux, but learns the stored quantity and the kernels while reconstructing the geometry.

As for the loss domain, RawNeRF [29] reweights a linear loss toward a tone-mapped error so that dark regions are not drowned out, and GS³ [4] anneals the gamma of its HDR targets toward linear radiance; we find the display domain to be the worse choice for forming the geometry of a relighting model (Sec. 4).

3. Light-Space Atlases

We reconstruct an object from images $\{Y_m\}$, each taken by a calibrated camera under a point light at $\mathbf{p}_m$ with RGB intensity $\mathbf{I}_m$ (Fig. 2).

3.1. Preliminaries

The scene is a set of 3D Gaussians with means μ_i , covariances Σ_i and opacities o_i [23], each carrying a base color $\mathbf{b}_i \in \mathbb{R}^3$ and a latent material code $\mathbf{f}_i \in \mathbb{R}^{32}$ whose first three channels define a learned unit normal $\mathbf{n}_i$. Splatting composites per-Gaussian attributes $\mathbf{c}_i$ front to back,

$$\mathbf{C}(u) = \sum_i w_i(u) \mathbf{c}_i, \quad w_i(u) = \alpha_i(u) \prod_{j < i} (1 - \alpha_j(u)), \quad (1)$$

where $\alpha_i(u)$ is the opacity of Gaussian i at pixel u . For a point $\mathbf{x}$ we write $\mathbf{l}, \mathbf{v}, \mathbf{h}$ for the unit light, view and half vectors and $E(\mathbf{x}) = \mathbf{I}/\|\mathbf{p} - \mathbf{x}\|^2$ for the inverse-square irradiance scale.

3.2. Splatting from the light deposits flux

Place a perspective camera at the light, aimed at the scene center, with a field of view that covers the splatted extent of 99.9% of the Gaussians. Its pixel u subtends a solid angle $d\omega_u$ and receives the flux $\mathbf{I} d\omega_u$. In the emission-absorption model underlying splatting, with its depth-sorted, EWA-projected primitives [23], Gaussian i absorbs the fraction $\alpha_i(u)$ of the light still travelling along the ray, so the flux it intercepts is

$$P_i(u) = \mathbf{I} d\omega_u w_i(u), \quad (2)$$

with the weights of Eq. (1). No normal or distance appears: since $E dA = \mathbf{I} \cos \theta dA/r^2 = \mathbf{I} d\omega$, foreshortening and falloff are accounted for by how many light pixels a surface covers. Rasterizing attributes from the light therefore averages them over the light absorbed along every light ray.¹

¹Pixels of a planar light image subtend solid angles that vary smoothly with the off-axis angle. The visibility statistics below are normalized by coverage and do not depend on this factor; we let the learned transfer term absorb it.

Atlas. Every Gaussian emits $\mathbf{c}_i = [\phi_i, z_i, z_i^2, 1]$, where z_i is its depth along the light’s axis relative to the scene center, normalized by the scene radius, and $\phi_i = \text{softplus}(\text{MLP}_\phi(\mathbf{f}_i, \mathbf{b}_i, \mathbf{n}_i \cdot \mathbf{l}_i)) \in \mathbb{R}_+^C$ are $C = 8$ learned flux features. One rasterization at 512^2 yields the atlas

$$\mathbf{A}(u) = \sum_i w_i(u) \mathbf{c}_i = [\Phi(u), M_1(u), M_2(u), M_0(u)], \quad (3)$$

where M_0 is the fraction of the pixel’s light absorbed by the scene, M_1/M_0 and M_2/M_0 are the first two moments of the depth at which it is absorbed, and Φ is a learned description of what it hit. A pyramid $\mathbf{A}_0, \dots, \mathbf{A}_{K-1}$ with $K = 7$ follows from repeated binomial blurring and $2\times$ decimation. Because every channel is linear in the weights w_i , each level still holds moments and flux sums of the deposit, over footprints growing from one texel to about a third of the object; this is what makes moment shadow maps filterable [13, 33].

3.3. Shading from the atlas

A receiver $\mathbf{x}$ projects into the light view at $u(\mathbf{x})$ with normalized depth $z(\mathbf{x})$ and samples every level bilinearly. At level k we form the mean depth $\mu_k = M_{1,k}/M_{0,k}$, a spread $s_k = (M_{2,k}/M_{0,k} - \mu_k^2 + \tau_k^2)^{1/2}$ that includes the texel size τ_k , the offset $\delta_k = z - \mu_k$ of the receiver behind the deposit, and $t_k = \delta_k/s_k$. If the depth of the deposit within a footprint is normally distributed, the fraction of light absorbed in front of the receiver is $M_{0,k} G(t_k)$, which gives the moment test

$$V_k = 1 - M_{0,k} G(t_k - \beta), \quad (4)$$

with the standard normal CDF G and a bias $\beta = 3$, measured in local spreads, against self-shadowing. The statistics $\mathbf{s}_k = [M_{0,k}, t_k, \delta_k, \log s_k, V_k]$ of all levels form the descriptor $\mathbf{S}(\mathbf{x}) \in \mathbb{R}^{5K}$.

Visibility. A residual network g , initialized to zero, refines the finest test,

$$V(\mathbf{x}) = \sigma(\text{logit } V_0(\mathbf{x}) + g(\mathbf{S}, \mathbf{f}, \mathbf{n} \cdot \mathbf{l})), \quad (5)$$

so training starts from a classical filtered shadow map and learns penumbras, grazing-angle bias and the partial transmittance of fur from the multi-scale statistics.

Transfer. Light entering a translucent object at $\mathbf{x}_i$ and leaving it at $\mathbf{x}_o$ contributes $\int R(\mathbf{x}_o, \mathbf{x}_i) E(\mathbf{x}_i) dA(\mathbf{x}_i)$, which by Eq. (2) is an integral over the light’s solid angle, $\mathbf{I} \int \sum_i R(\mathbf{x}_o, \mathbf{x}_i) w_i(u) d\omega$, of a receiver-dependent kernel against the deposit. Translucent shadow maps evaluate such integrals with stored irradiance and fixed diffusion kernels [7, 11]; we learn both the deposited quantity and

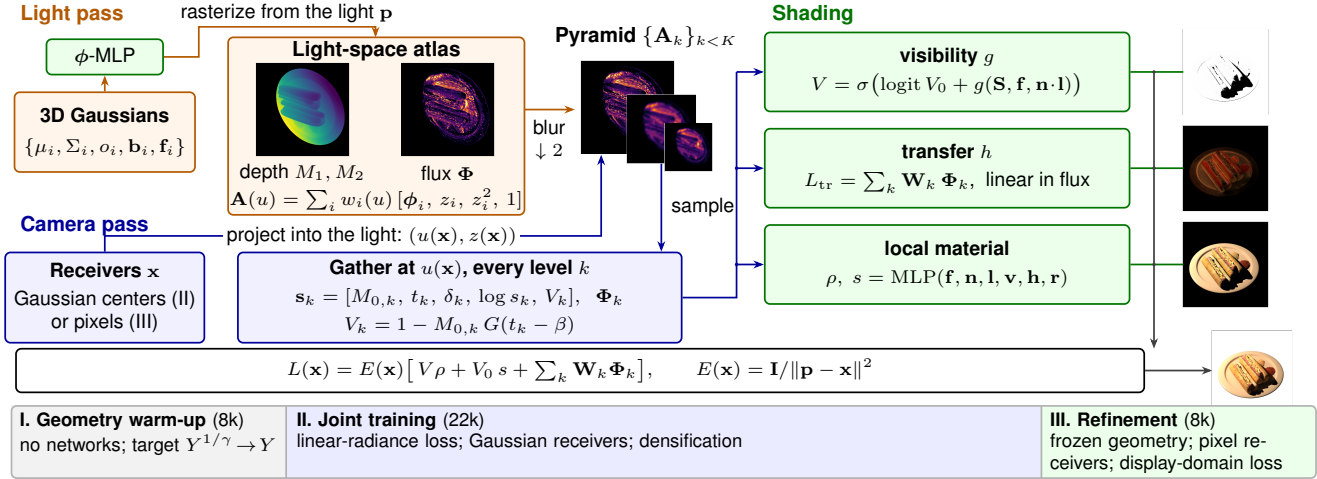


Figure 2. **Overview.** For every training or test light, the Gaussians are rasterized once from the light. Each Gaussian deposits learned flux features ϕ_i and its depth moments with exactly the weight of the light it intercepts (Eq. (3)). A receiver projects into the light, gathers multi-scale statistics and flux from the atlas pyramid, and is shaded by three small networks: a residual on the moment shadow test, a transfer kernel that is linear in the gathered flux, and a local material. Thumbnails are actual buffers of our Hotdog model on a test view (from top: visibility V , transfer radiance $E L_{tr}$, local radiance $E(V\rho + V_0s)$). Bottom: the three-stage schedule of Sec. 4.

the kernel, using the pyramid levels as kernels of increasing width:

$$L_{tr}(\mathbf{x}) = \sum_k \mathbf{W}_k \Phi_k(\mathbf{x}), \quad \{\mathbf{W}_k\}_k = \text{softplus}(h(\mathbf{S}, \mathbf{f}, \mathbf{n} \cdot \mathbf{l}, \mathbf{n} \cdot \mathbf{v})), \quad (6)$$

with $\mathbf{W}_k \in \mathbb{R}_+^{3 \times C}$. The term is linear in the deposited flux, and the offsets δ_k tell the kernel how much material the light crossed; a sum-of-Gaussians translucent shadow map is the special case of one flux channel and fixed weights that decay with δ_k .

Local material and radiance. The local reflectance $\rho(\mathbf{x})$ is a latent-code neural BRDF, an MLP of $\mathbf{f}$, $\mathbf{n}$, Fourier encodings of $\mathbf{l}$, $\mathbf{v}$, $\mathbf{h}$ and the reflected view direction, their cosines, and highlight hints $e^{\kappa(\mathbf{n} \cdot \mathbf{h} - 1)}$ in the spirit of NRHints [45], multiplied by a smoothed cosine $\psi(\mathbf{n} \cdot \mathbf{l})$ with $\psi(c) = \text{softplus}(20c)/20$. Position enters only through eight light-independent coefficients that mix eight light-dependent responses, which limits how much cast shadow the network can bake into position. A bank of isotropic lobes $s(\mathbf{x})$ with fixed sharpness $\kappa \in \{8, \dots, 2048\}$ and light-independent weights carries sharp highlights. The outgoing radiance is

$$L(\mathbf{x}) = E(\mathbf{x})[V(\mathbf{x})\rho(\mathbf{x}) + V_0(\mathbf{x})s(\mathbf{x}) + L_{tr}(\mathbf{x})]. \quad (7)$$

The lobes are shadowed by the geometric test V_0 , so highlights keep a gradient path where the learned V is still wrong. Factoring $E(\mathbf{x})$ out of every term lets the networks predict reflectance-like quantities; for L_{tr} this approximates the falloff of nearby deposits by that of the receiver, which holds when the light is far compared with the gather footprint.

All inputs of g and h are relative to the light, so the operator moves with the light rather than being indexed by its position.

3.4. Receivers: Gaussians or pixels

Equation (7) can be evaluated at two kinds of receivers. *Gaussian receivers* shade every Gaussian center μ_i and composite the colors with Eq. (1); *pixel receivers* composite $\mathbf{b}$, $\mathbf{f}$ and the expected depth first, lift each pixel to a world point and shade it once. Gaussian receivers give the geometry the ordinary rasterization gradients that densification relies on and keep overlapping surfaces from mixing their materials, but they inherit the per-primitive shadow limitation discussed in Sec. 1. Pixel receivers evaluate the shadow test at each pixel’s own surface point, so a shadow boundary can cross a large Gaussian, and they resolve appearance at image resolution. LiSA therefore uses Gaussian receivers while the geometry is optimized and pixel receivers once it is fixed (Sec. 4); the atlas is the same in both cases.

4. Radiometrically Staged Optimization

4.1. Where the loss is computed

Photometric losses for relighting are commonly computed on display-encoded images, the native format of LDR captures. For an encoding $f(L) = L^{1/\gamma}$ with $\gamma = 2.2$, the gradient of an ℓ_1 loss with respect to a predicted radiance $\hat{L}$ is scaled by

$$f'(\hat{L}) = \frac{1}{\gamma} \hat{L}^{1/\gamma - 1}, \quad (8)$$

which grows without bound as $\hat{L} \rightarrow 0$: a pixel predicted at one percent of white receives twelve times the gradient

of a white one. In Eq. (7) the prediction is a product of learned factors, and early in training a pixel is often dark only because its visibility, normal or material is still wrong. We hypothesize that the display-domain weighting lets these errors dominate the gradients that reach the geometry, which the optimizer resolves by smoothing thin structures into blobs. The controlled experiment of Sec. 5.4 confirms that the loss domain decides the outcome, although it does not isolate the mechanism.

4.2. Three stages

We seed 20k Gaussians uniformly inside the visual hull of 32 training silhouettes (a point is kept if it falls inside at least 90% of them) and initialize normals to point away from the scene center.

I. Geometry warm-up (8k iterations). A network-free point-light model colors every Gaussian with $\text{softplus}(\mathbf{b}_i) E(\mu_i) \psi(\mathbf{n}_i \cdot \mathbf{l}_i)$, and only geometry, base colors and normals are optimized. The renderer output is linear while the target moves: it is the display-encoded image $Y^{1/\gamma}$ for the first two thirds of the warm-up, after which its exponent is annealed linearly from $1/\gamma$ to 1. The geometry thus forms against a target with compressed dynamic range and is carried continuously into linear radiance. GS³ applies a comparable annealing to its HDR training images [4]; we apply it to every data family and, more importantly, keep the joint training that follows in linear radiance.

II. Joint training (22k iterations). The full model of Eq. (7) is trained with Gaussian receivers and a linear-radiance loss; densification and pruning follow 3DGS [23] until iteration 25k, with at most 400k Gaussians.

III. Appearance refinement (8k iterations). Geometry and camera calibration are frozen, receivers switch to pixels, and the loss returns to the display domain in which images are evaluated. Once the geometry can no longer move, the extra weight on dark pixels stops being harmful; in our experiments it improves all three metrics for both kinds of receivers (Tab. 5).

All stages minimize $0.8 \ell_1 + 0.2 (1 - \text{SSIM}) + 0.5 \|\hat{\alpha} - \alpha\|_1$ with the foreground mask α , plus a weak ℓ_2 penalty on the codes $\mathbf{f}$.

4.3. Implementation details

LiSA is implemented in PyTorch with the gsplat rasterizer [44]. The atlas has 512^2 texels, $K = 7$ levels and $C = 8$ flux channels; the material and transfer MLPs have width 128 and the visibility MLP width 64, with outputs initialized so that training starts from $L_{\text{tr}} \approx 0$ and $V = V_0$. Networks use Adam with a learning rate of 10^{-3} decayed to about a quarter; Gaussians use the learning rates of 3DGS. On real

captures we also refine the rotation of every training camera about its calibrated center and, after each step, remove the global translation that such refinements can absorb, so that the reconstruction stays in the calibrated frame; test cameras and lights are never adjusted. One configuration serves all 18 scenes, with only the data conventions of each benchmark differing, and a complete run takes 17 to 28 minutes (21 on average) on one NVIDIA RTX 6000 Ada.

5. Experiments

5.1. Setup

Data. We use every scene of three public OLAT benchmarks: the seven real captures of NRHints [45] (522–1500 training and 66–200 test images each), the six HDR synthetic scenes of GS³ [4] (2000/400), and the five subsurface-scattering scenes of SSS-GS [12] (500/500). We keep each benchmark’s conventions (512 px images, 256 px for SSS-GS; white background and $\gamma = 2.2$ display for the HDR scenes) and use all official training images and all 5,691 official test images.

Baselines. We compare with the four recent Gaussian methods that train per scene from point-light captures and provide code, GS³ [4], SSD-GS [49], OLAT Gaussians (OLAT-GS) [25] and RNG [14], retrained on all 18 scenes with their released code and default schedules of 60k–100k iterations (ours: 38k). On real captures, GS³, SSD-GS and OLAT-GS refine training poses as in their code; no method, including ours, fits test cameras or lights. Published numbers are not comparable, as protocols differ in resolution, training images and test-pose fitting. **[Optional, E5] Add SSS-GS [12] on its own scenes once a stable reproduction under this protocol exists (our runs diverged on three of five scenes).**

Metrics. We report PSNR, SSIM [41] and VGG-LPIPS [46] on full images, averaged over the test views of each scene and then over scenes. All predictions are scored against one common set of 8-bit targets, which changed the baselines’ own averages by at most 0.005 dB and no ranking. Main results use one run per method and scene; ablations use three seeds.

5.2. Comparison with the state of the art

LiSA has the best average PSNR, SSIM and LPIPS over the 18 scenes (Tab. 1), 1.89 dB ahead of the next best method, and the best PSNR on 9 scenes; it beats SSD-GS, GS³ and OLAT-GS individually on 13, 13 and 12 scenes. On the GS³ scenes the largest margins occur where appearance is governed by moving cast shadows and by light passing through translucent objects, with gains of 1.90 dB on Hotdog and 1.91 dB on Translucent over the best baseline.

Table 1. **Comparison on all 18 scenes of three OLAT benchmarks.** Scene-averaged metrics over all 5,691 official test images, every method scored against identical targets at the original test calibration without test-time fitting. Best in **bold**, second underlined. Training iterations: RNG 30k+70k, OLAT-GS 30k+30k, GS³ 100k, SSD-GS 100k (60k on SSS), LiSA 38k. Per-scene results are in the supplement.

Method	NRHints, real (7)			GS ³ , synthetic (6)			SSS-GS, synthetic (5)			All scenes (18)		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
RNG [14]	20.94	.828	.203	22.14	.895	.132	37.10	.983	.028	25.83	.893	.131
OLAT-GS [25]	27.26	.886	.114	27.11	.940	.079	40.74	.990	.016	30.95	.933	.075
GS ³ [4]	27.28	.884	.116	31.97	.967	.049	36.81	.983	.028	31.49	.939	.069
SSD-GS [49]	27.20	.882	.117	31.53	.965	.050	37.79	.986	.023	31.59	.939	.068
LiSA (ours)	29.28	.909	.100	32.40	.970	.038	<u>40.63</u>	.990	.016	33.47	.952	.056

Figure 3 shows why: the shadows of the vase handle and of the hot dogs fall across large floor and plate Gaussians, which per-primitive visibility approximates by fragmenting them, whereas the pixel-wise atlas test reproduces their outlines. LiSA is behind on Drums (−1.09 dB) and AnisoMetal (−0.85 dB), whose thin or anisotropic highlights our isotropic material misses (Sec. 5.6). On the subsurface-scattering scenes, LiSA and OLAT-GS are on par (40.63 vs. 40.74 dB, LPIPS slightly in our favor) and both lead the others by at least 2.8 dB, although OLAT-GS needs a separate surface reconstruction stage.

On real captures, LiSA leads on four of seven scenes and by 2.0 dB on average, most on Cat, Fish and Pixiu, and trails the best method by 0.8–1.6 dB on CatSmall, CupFabric and FurScene. Fixed-calibration scores on these captures depend on registration as well as appearance: the official poses carry small errors, and refining training poses without anchoring the reconstruction can shift it relative to the test cameras. Our gauge-constrained refinement (Sec. 4.3) avoids such shifts by construction, so part of our margin on Cat, Fish and Pixiu likely reflects registration rather than shading. **[TO RUN] E2: LiSA without pose refinement; all methods after per-view 2D alignment (radius 24 px), reported next to the fixed-calibration numbers.**

5.3. What the atlas contributes

Table 2 ablates the atlas on five scenes covering opaque, metallic, furry, translucent and subsurface-scattering materials; the full model varies by 0.11 dB across seeds. Using only the finest atlas level costs 0.44 dB and replacing the learned visibility by the plain moment test of Eq. (4) costs 0.24 dB (single seeds, to be read against the 0.11 dB seed-to-seed deviation), so both the multi-scale statistics and the refinement matter.

Removing the transfer term is the most damaging change (−1.37 dB, LPIPS 0.047→0.055 on every seed), and the damage follows the material (Tab. 3). It is negligible on Drums and FurScene, 1.42 dB on AnisoMetal, and 2.29 and 2.99 dB on Translucent and Soap. The model itself routes 6–7% of its linear radiance through the transfer term on the

Table 2. **Ablation of the atlas** on five scenes (Drums, AnisoMetal, Translucent, FurScene, Soap), scene-averaged. For rows with three seeds we report the mean over seeds and the standard deviation of PSNR; single-seed rows compare to the seed-0 full model (31.53 dB). [†]Capacity-matched MLP (49.1k vs. 49.6k parameters) that reads the receiver’s code, position and light/view directions but not the atlas; shadows still come from the atlas.

Variant	Seeds	PSNR↑	SSIM↑	LPIPS↓
Full model	3	31.51±0.11	.954	.047
w/o transfer term	3	30.15±0.11	.948	.055
transfer → local residual [†]	3	31.54±0.03	.954	.047
single atlas level ($K=1$)	1	31.09	.952	.049
moment test only ($g \equiv 0$)	1	31.29	.952	.048
per-Gaussian visibility [‡]	3	TBD	TBD	TBD

[‡]**[TO RUN] E3: per-Gaussian splatted visibility as in GS³/SSD-GS; atlas kept for transfer.**

Table 3. **Where the transfer term matters.** Per-scene PSNR differences (mean±std of paired differences over three seeds) of the full model against removing the transfer term and against the capacity-matched local residual, and the share of linear radiance the full model routes through the transfer term (mean over 8 test views).

Scene	vs. w/o transfer	vs. local residual	Transfer share
Drums	+0.10±0.02	−0.09±0.02	6%
FurScene	+0.02±0.03	+0.02±0.02	7%
AnisoMetal	+1.42±0.69	−0.18±0.75	7%
Translucent	+2.29±0.23	+0.41±0.26	17%
Soap	+2.99±0.30	−0.29±0.11	58%

first three scenes but 17% on Translucent and 58% on Soap, and over all 18 scenes the share is 6–9% on the opaque synthetic objects and 20–58% on the subsurface-scattering ones (supplement).

Is it the atlas that matters, or just an additive radiance path? A capacity-matched MLP that sees the receiver’s code, position and light and view directions but not the atlas, with shadows still from the atlas, performs on par on

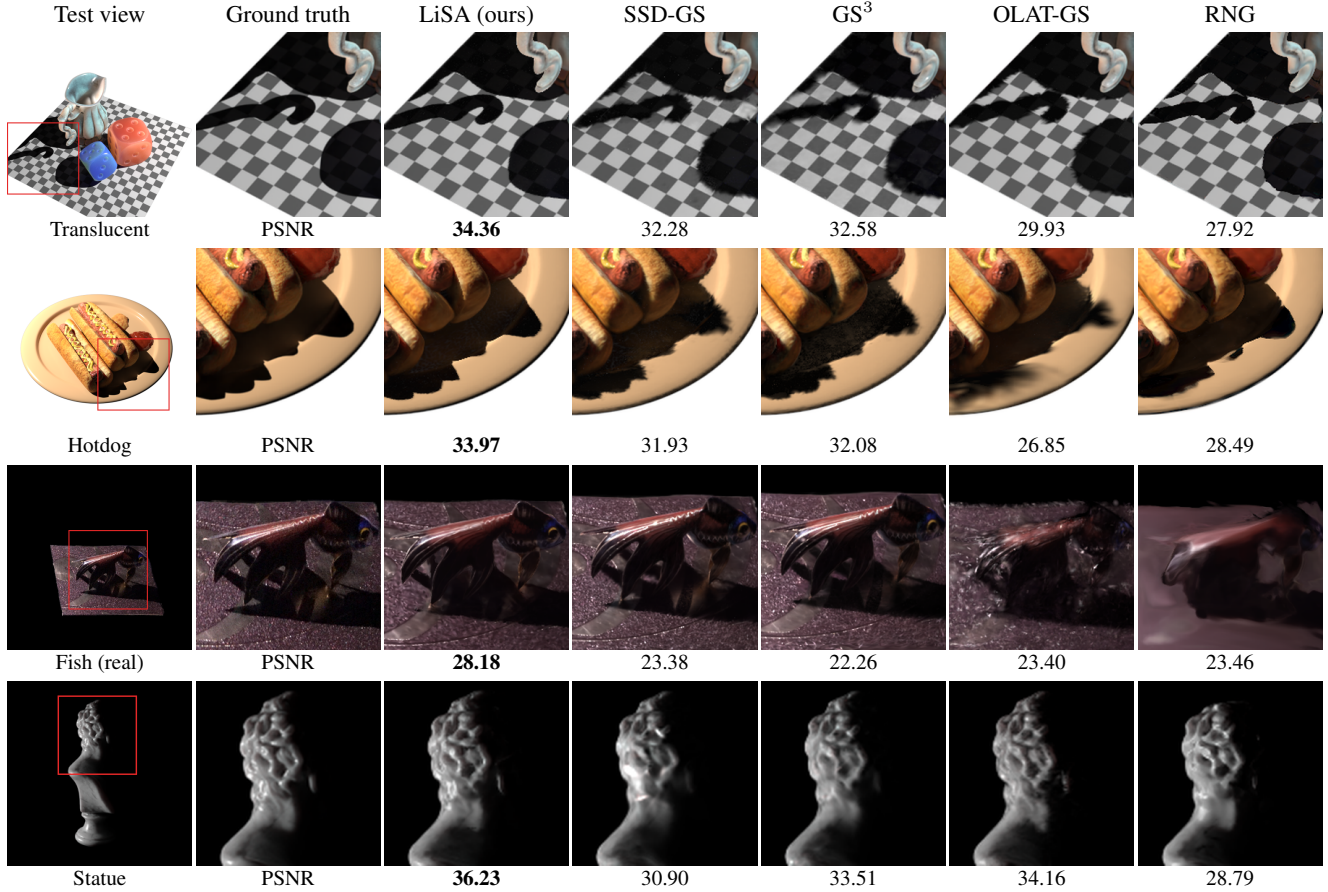


Figure 3. **Qualitative comparison** on test views (crops of the boxed region; PSNR of the full view). Per-primitive shadows of SSD-GS and GS³ fragment the moving shadows of the vase handle and the hot dogs, while the atlas reproduces their outlines. Views lie at the median of LiSA’s per-view margin over the best baseline (75th percentile for Statue); Fish is a real capture (Sec. 5.2).

average (31.54 vs. 31.51 dB). On most materials, the transfer term therefore mainly supplies an unshadowed, nonnegative radiance path that a local network can also provide. Reading the atlas wins on every seed only on Translucent (+0.41 dB), the one scene where light visibly crosses the object before reaching the visible surface, while the local residual wins on every seed on Soap (−0.29 dB) and marginally on Drums; on AnisoMetal the full model is also less stable across seeds (26.4–27.9 dB). We therefore do not credit the transfer term with our overall margin. The evidence points instead to the light-space visibility and the optimization schedule, a hypothesis that E3 in Tab. 2 tests directly. What favors the light-relative design should show under light extrapolation, which the benchmark test splits do not probe: their test lights lie a median of 1.1° from the nearest training light. **[TO RUN] E1 (Tab. 4): retrain all methods and the local residual without the views whose lights fall in held-out 30° angular groups; evaluate on those views. Hypothesis to verify: the atlas degrades less than the local residual.**

Table 4. **Held-out light directions ([TO RUN] E1)**. PSNR on training-split views whose lights fall in held-out 30° angular groups; every method is retrained without those views.

Method	AnisoMetal	Translucent	Bunny	Dragon	Pixiu
GS ³	TBD	TBD	TBD	TBD	TBD
SSD-GS	TBD	TBD	TBD	TBD	TBD
OLAT-GS	TBD	TBD	TBD	TBD	TBD
LiSA, local residual	TBD	TBD	TBD	TBD	TBD
LiSA (ours)	TBD	TBD	TBD	TBD	TBD

5.4. What the optimization contributes

Loss domain. Two 16k-iteration runs on Lego and Drums share the initialization, schedule, warm-up and random states and differ only in the domain of the loss; we evaluate them on 200 views held out from the training set. With the display-domain loss both geometries collapse into blobs (Fig. 4); with the radiometric schedule, PSNR rises from 23.02 to 30.25 dB on Lego and from 24.10 to 30.10 dB on Drums

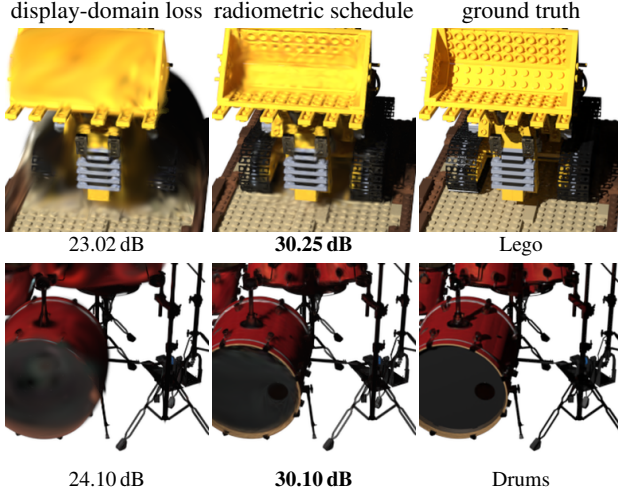


Figure 4. **The loss domain decides the geometry.** Two runs that share initialization, schedule, random state and every other setting, differing only in whether the loss is computed on display-encoded or (after the annealed warm-up) linear radiance; 16k iterations, PSNR over 200 held-out training views. All 200 views improve on both scenes.

Table 5. **Appearance refinement** (stage III, 8k iterations on the same frozen 30k geometry), averaged over Lego, Drums and Soap. Our default refines pixels in the display domain.

Receivers	Loss domain	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Gaussians	linear	34.04	.971	.036
Gaussians	display	34.24	<u>.974</u>	.033
pixels	linear	33.77	.973	<u>.032</u>
pixels	display	<u>34.05</u>	.976	.029

(LPIPS 0.122 $\rightarrow$ 0.057 and 0.060 $\rightarrow$ 0.034), and every held-out view improves, by a median of 7.2 and 6.0 dB. **[TO RUN] E4: full-protocol ablation on the five scenes of Tab. 2: (a) display-domain loss throughout, (b) no warm-up, (c) pixel receivers during joint training, (d) no stage III.**

Receivers and refinement. On frozen 30k geometry (Tab. 5), refining in the display domain improves all three metrics for both kinds of receivers. Pixel receivers give the best SSIM and LPIPS and Gaussian receivers a 0.19 dB higher PSNR; we keep pixels for their perceptual quality.

5.5. Cost

On three scenes profiled on an otherwise idle GPU (Tab. 6), LiSA trains 1.6–5.5 times faster than GS³ and SSD-GS and uses 228k Gaussians on Drums against about 550k for both. Over all 18 scenes of our runs, mean training time is 21 minutes for LiSA against 55 for GS³, 68 for SSD-GS (14 scenes with logs), 69 for OLAT-GS and 107 for RNG, although

Table 6. **Cost** on three profiled scenes, each run alone on one RTX 6000 Ada: training time in minutes (whole process, both LiSA stages) and rendering frame rate with a new light in every frame, including the light pass of LiSA.

Method	Drums		FurScene		Soap	
	min	FPS	min	FPS	min	FPS
GS ³	73.5	61	51.0	191	39.4	228
SSD-GS	141.1	29	61.2	136	26.8	198
LiSA	25.7	63	19.9	74	16.7	104

these runs were not timed under strictly uniform conditions. The extra light pass keeps rendering interactive at 63–104 frames per second with a new light in every frame, but on small scenes it makes LiSA slower than GS³ and SSD-GS.

5.6. Limitations

Isotropic highlight lobes cannot represent brushed metal and miss thin specular streaks: on Drums, 0.27% of the pixels, flagged as narrow highlights in the ground truth, account for 72% of our excess squared error relative to GS³. A single perspective atlas assumes the light lies outside the object; lights inside the scene would need cube or cascaded atlases. Light that reaches a receiver only through regions the light cannot see, such as multi-bounce interreflection, is beyond one light-space gather, and the local/transfer split is a learned approximation rather than a physical decomposition. The linear-domain schedule is not uniformly beneficial: an earlier display-domain variant of our model, which also differed in other choices, scored 0.40 dB higher on the subsurface-scattering scenes and 0.22 dB higher on real captures while trailing by 3.76 dB on the GS³ scenes (supplement). Finally, the benchmark test lights are close to training lights, so extrapolation to distant light positions remains to be shown (E1).

6. Conclusion

For a point light, rasterizing a Gaussian scene from the light is more than a shadow map: its compositing weights are the flux each primitive intercepts. LiSA turns this into a learned light-space atlas that gives every receiver a filtered, refinable shadow test and a transfer term linear in the surrounding deposit, for one extra rasterization per light. Where the loss is computed proved as important as the representation, and together they give the best average accuracy on 18 scenes of three benchmarks at a fraction of the usual training time. Our analysis also delimits the contribution: on most materials we tested, a local network matches the atlas transfer. Several point lights superpose with one atlas each; lights inside the scene, anisotropic highlights and held-out light directions, the regime the light-relative design was built for, are the next steps.

References

- [1] Thomas Annen, Tom Mertens, Hans-Peter Seidel, Eddy Flerackers, and Jan Kautz. Exponential shadow maps. In *Proceedings of Graphics Interface*, 2008. 2
- [2] Sai Bi, Zexiang Xu, Pratul Srinivasan, Ben Mildenhall, Kalyan Sunkavalli, Miloš Hašan, Yannick Hold-Geoffroy, David Kriegman, and Ravi Ramamoorthi. Neural reflectance fields for appearance acquisition. *arXiv preprint arXiv:2008.03824*, 2020. 1, 2
- [3] Sai Bi, Zexiang Xu, Kalyan Sunkavalli, Miloš Hašan, Yannick Hold-Geoffroy, David Kriegman, and Ravi Ramamoorthi. Deep reflectance volumes: Relightable reconstructions from multi-view photometric images. In *European Conference on Computer Vision (ECCV)*, 2020. 2
- [4] Zoubin Bi, Yixin Zeng, Chong Zeng, Fan Pei, Xiang Feng, Kun Zhou, and Hongzhi Wu. GS³: Efficient relighting with triple Gaussian splatting. In *SIGGRAPH Asia Conference Papers*, 2024. 1, 2, 3, 5, 6
- [5] Hongze Chen, Zehong Lin, and Jun Zhang. GI-GS: Global illumination decomposition on Gaussian splatting for inverse rendering. In *International Conference on Learning Representations (ICLR)*, 2025. 2
- [6] Cyril Crassin, Fabrice Neyret, Miguel Sainz, Simon Green, and Elmar Eisemann. Interactive indirect illumination using voxel cone tracing. *Computer Graphics Forum*, 30(7), 2011. 3
- [7] Carsten Dachsbacher and Marc Stamminger. Translucent shadow maps. In *Eurographics Symposium on Rendering (EGSR)*, 2003. 2, 3
- [8] Carsten Dachsbacher and Marc Stamminger. Reflective shadow maps. In *Proceedings of the Symposium on Interactive 3D Graphics and Games (I3D)*, 2005. 2
- [9] Sayantan Datta, Derek Nowrouzezahrai, Christoph Schied, and Zhao Dong. Neural shadow mapping. In *ACM SIGGRAPH Conference Proceedings*, 2022. 3
- [10] Paul Debevec, Tim Hawkins, Chris Tchou, Haarm-Pieter Duiker, Westley Sarokin, and Mark Sagar. Acquiring the reflectance field of a human face. In *Proceedings of SIGGRAPH*, pages 145–156, 2000. 1
- [11] Eugene d'Eon, David Luebke, and Eric Enderton. Efficient rendering of human skin. In *Eurographics Symposium on Rendering (EGSR)*, 2007. 2, 3
- [12] Jan-Niklas Dihlmann, Arjun Majumdar, Andreas Engelhardt, Raphael Braun, and Hendrik P. A. Lensch. Subsurface scattering for Gaussian splatting. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 121765–121789, 2024. 1, 2, 5
- [13] William Donnelly and Andrew Lauritzen. Variance shadow maps. In *Proceedings of the Symposium on Interactive 3D Graphics and Games (I3D)*, 2006. 2, 3
- [14] Jiahui Fan, Fujun Luan, Jian Yang, Miloš Hašan, and Beibei Wang. RNG: Relightable neural Gaussians. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 1, 2, 5, 6
- [15] Duan Gao, Guojun Chen, Yue Dong, Pieter Peers, Kun Xu, and Xin Tong. Deferred neural lighting: Free-viewpoint relighting from unstructured photographs. *ACM Transactions on Graphics*, 39(6), 2020. 3
- [16] Jian Gao, Chun Gu, Youtian Lin, Zhihao Li, Hao Zhu, Xun Cao, Li Zhang, and Yao Yao. Relightable 3D Gaussians: Realistic point cloud relighting with BRDF decomposition and ray tracing. In *European Conference on Computer Vision (ECCV)*, 2024. 2
- [17] Chun Gu, Xiaofei Wei, Zixuan Zeng, Yuxuan Yao, and Li Zhang. IRGS: Inter-reflective Gaussian splatting with 2D Gaussian ray tracing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 2
- [18] Henrik Wann Jensen, Stephen R. Marschner, Marc Levoy, and Pat Hanrahan. A practical model for subsurface light transport. In *Proceedings of SIGGRAPH*, 2001. 2
- [19] Kaiwen Jiang, Jia-Mu Sun, Zilu Li, Dan Wang, Tzu-Mao Li, and Ravi Ramamoorthi. Differentiable light transport with Gaussian surfels via adapted radiosity for efficient relighting and geometry reconstruction. *ACM Transactions on Graphics (Proc. SIGGRAPH Asia)*, 2025. 2
- [20] Yingwenqi Jiang, Jiadong Tu, Yuan Liu, Xifeng Gao, Xiaoxiao Long, Wenping Wang, and Yuexin Ma. GaussianShader: 3D Gaussian splatting with shading functions for reflective surfaces. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2
- [21] Haian Jin, Isabella Liu, Peijia Xu, Xiaoshuai Zhang, Songfang Han, Sai Bi, Xiaowei Zhou, Zexiang Xu, and Hao Su. TensorIR: Tensorial inverse rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2
- [22] Anton Kaplanyan and Carsten Dachsbacher. Cascaded light propagation volumes for real-time indirect illumination. In *Proceedings of the Symposium on Interactive 3D Graphics and Games (I3D)*, 2010. 3
- [23] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4), 2023. 1, 3, 5
- [24] Tae-Yong Kim and Ulrich Neumann. Opacity shadow maps. In *Eurographics Workshop on Rendering Techniques*, 2001. 2
- [25] Zhiyi Kuang, Yanchao Yang, Siyan Dong, Jiayue Ma, Hongbo Fu, and Youyi Zheng. OLAT Gaussians for generic relightable appearance acquisition. In *SIGGRAPH Asia Conference Papers*, pages 1–11, 2024. 1, 2, 5, 6
- [26] Zhihao Liang, Qi Zhang, Ying Feng, Ying Shan, and Kui Jia. GS-IR: 3D Gaussian splatting for inverse rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2
- [27] Zhenyuan Liu, Yu Guo, Xinyuan Li, Bernd Bickel, and Ran Zhang. BiGS: Bidirectional Gaussian primitives for relightable 3D Gaussian splatting. In *International Conference on 3D Vision (3DV)*, 2025. 1, 2
- [28] Tom Lokovic and Eric Veach. Deep shadow maps. In *Proceedings of SIGGRAPH*, 2000. 2
- [29] Ben Mildenhall, Peter Hedman, Ricardo Martin-Brualla, Pratul P. Srinivasan, and Jonathan T. Barron. NeRF in the dark: High dynamic range view synthesis from noisy raw

- images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. 3
- [30] Aymen Mir, Riza Alp Guler, Jian Wang, Gerard Pons-Moll, and Bing Zhou. Animated 3DGS avatars in diverse scenes with consistent lighting and shadows. *arXiv preprint arXiv:2601.01660*, 2026. 2
- [31] Nicolas Moenne-Loccoz, Ashkan Mirzaei, Or Perel, Riccardo de Lutio, Janick Martinez Esturo, Gavriel State, Sanja Fidler, Nicholas Sharp, and Zan Gojcic. 3D Gaussian ray tracing: Fast tracing of particle scenes. *ACM Transactions on Graphics*, 43(6), 2024. 2
- [32] Oliver Nalbach, Elena Arabadzhiyska, Dushyant Mehta, Hans-Peter Seidel, and Tobias Ritschel. Deep shading: Convolutional neural networks for screen space shading. *Computer Graphics Forum*, 36(4), 2017. 3
- [33] Christoph Peters and Reinhard Klein. Moment shadow mapping. In *Proceedings of the ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D)*, 2015. 2, 3
- [34] William T. Reeves, David H. Salesin, and Robert L. Cook. Rendering antialiased shadows with depth maps. In *Proceedings of SIGGRAPH*, pages 283–291, 1987. 2
- [35] Shunsuke Saito, Gabriel Schwartz, Tomas Simon, Junxuan Li, and Giljoo Nam. Relightable Gaussian codec avatars. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2
- [36] Peter-Pike Sloan, Jan Kautz, and John Snyder. Precomputed radiance transfer for real-time rendering in dynamic, low-frequency lighting environments. *ACM Transactions on Graphics*, 21(3), 2002. 2
- [37] Pratul P. Srinivasan, Boyang Deng, Xiuming Zhang, Matthew Tancik, Ben Mildenhall, and Jonathan T. Barron. NeRV: Neural reflectance and visibility fields for relighting and view synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 2
- [38] Justus Thies, Michael Zollhöfer, and Matthias Nießner. Deferred neural rendering: Image synthesis using neural textures. *ACM Transactions on Graphics*, 38(4), 2019. 3
- [39] Peng Wang, Lingjie Liu, Yuan Liu, Christian Theobalt, Taku Komura, and Wenping Wang. NeuS: Learning neural implicit surfaces by volume rendering for multi-view reconstruction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. 2
- [40] Shaofei Wang, Tomas Simon, Igor Santesteban, Timur Bagautdinov, Junxuan Li, Vasu Agrawal, Fabian Prada, Shoou-I Yu, Pace Nalbone, Matt Gramlich, Roman Lubachersky, Chenglei Wu, Javier Romero, Jason Saragih, Michael Zollhoefer, Andreas Geiger, Siyu Tang, and Shunsuke Saito. Relightable full-body Gaussian codec avatars. In *ACM SIGGRAPH Conference Papers*, 2025. 2
- [41] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004. 5
- [42] Lance Williams. Casting curved shadows on curved surfaces. In *Proceedings of SIGGRAPH*, pages 270–274, 1978. 2
- [43] Peiyu Xu, Xin Sun, Krishna Mullia, Raymond Fei, Iliyan Georgiev, and Shuang Zhao. Stochastic ray tracing for the reconstruction of 3D Gaussian splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. 2
- [44] Vickie Ye, Ruilong Li, Justin Kerr, Matias Turkulainen, Brent Yi, Zhuoyang Pan, Otto Seiskari, Jianbo Ye, Jeffrey Hu, Matthew Tancik, and Angjoo Kanazawa. gsplat: An open-source library for Gaussian splatting. *Journal of Machine Learning Research*, 26(34):1–17, 2025. 5
- [45] Chong Zeng, Guojun Chen, Yue Dong, Pieter Peers, Hongzhi Wu, and Xin Tong. Relighting neural radiance fields with shadow and highlight hints. In *ACM SIGGRAPH Conference Proceedings*, 2023. 1, 2, 4, 5
- [46] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. 5
- [47] Xiuming Zhang, Sean Fanello, Yun-Ta Tsai, Tiancheng Sun, Tianfan Xue, Rohit Pandey, Sergio Orts-Escolano, Philip Davidson, Christoph Rhemann, Paul Debevec, Jonathan T. Barron, Ravi Ramamoorthi, and William T. Freeman. Neural light transport for relighting and view synthesis. *ACM Transactions on Graphics*, 40(1), 2021. 3
- [48] Xiuming Zhang, Pratul P. Srinivasan, Boyang Deng, Paul Debevec, William T. Freeman, and Jonathan T. Barron. NeRFactor: Neural factorization of shape and reflectance under an unknown illumination. *ACM Transactions on Graphics*, 40(6), 2021. 2
- [49] Iris Zheng, Guojun Tang, Alexander Doronin, Paul Teal, and Fang-Lue Zhang. SSD-GS: Scattering and shadow decomposition for relightable 3D Gaussian splatting. In *International Conference on Learning Representations (ICLR)*, 2026. 1, 2, 5, 6